

INTERNSHIP SUMMARY REPORT

on the activities carried out during the internship

Host organisation S.A. "Orange Systems" _____
Faculty of *Computers, Informatics, and Microelectronics* _____
Study programme *Software Engineering* _____
Mode of study *full time* _____
Year of study: 3 _____ **Group:** FAF-243 _____
Internship period: from 01.09.2026 _____ to 26.09.2026 _____
Activities carried out and results obtained:

Activities carried out:

During my internship at Orange Systems, I worked with my team on SecretShare, a web application for sharing confidential information with a specific recipient for a single retrieval. My main contribution focused on backend development, troubleshooting, and improving the reliability of the application. I centralised the application configuration in one settings class, introduced configurable secret lifetimes, and improved input validation and error handling. I also automated database migrations so that the required database structure would be prepared when the application started. Another part of my work involved investigating problems with secret retrieval, rate limiting, and the deployment process. For these changes, I added or updated automated tests to check both normal behaviour and exceptional situations. Alongside development, I updated the project documentation and helped align the final report with the university guidelines and the actual implementation. This involved reviewing technical descriptions, diagrams, references, and formatting.

Results obtained:

My work made the backend easier to configure and helped resolve several issues affecting its reliability. Secrets could be stored with configurable expiration times, invalid requests received clearer responses, and database migrations became part of the startup process.

An important result was correcting the retrieval mechanism so that an unauthorised user could no longer cause a secret to be deleted. Tests with simultaneous requests also checked that the secret could be retrieved only once and only by its intended recipient. I corrected client IP handling for rate limiting and audit records, and fixed the preview deployment workflow so that incomplete tests would cause the deployment check to fail.

On a personal level, I gained practical experience in investigating unfamiliar code, writing regression tests, and following a problem across several parts of an application.

Difficulties encountered during the internship:

The most challenging issue concerned the order in which the application checked the recipient and deleted a secret. Initially, an unauthorised request was rejected, but the secret was deleted anyway. Simply checking the recipient before deleting the secret would have introduced another problem: two simultaneous requests could potentially retrieve the same information. I solved this by combining the recipient check and deletion in one atomic Redis operation, then testing it with competing requests. This task helped me understand why a solution must remain correct under concurrent use.

Another unexpected difficulty was discovering that successful preview deployment checks did not mean that the tests had actually run. A command consumed the remaining deployment instructions, allowing the process to finish early without reporting a failure. Investigating this taught me to examine logs and verify that the expected checks had completed.

I also encountered differences between local and deployed behaviour. Behind the proxy, all users appeared to have the same IP address and therefore shared one rate limit. Resolving this required understanding how requests passed through the infrastructure. These situations encouraged a more patient and systematic approach to debugging.

Conclusions and recommendations:

The internship helped me connect university knowledge with the practical demands of developing an application in a team. I learned that completing a feature also involves checking its behaviour in the deployed environment, handling unexpected situations, and keeping tests and documentation consistent with the code.

The most valuable part of the experience was investigating problems whose causes were not immediately obvious. Although this could be frustrating at first, reproducing an issue and gradually narrowing down its cause made the work rewarding. I became more confident in reading existing code, questioning assumptions, and explaining why a particular solution was necessary.

For future internships, I would recommend preparing access to the development and testing environments early and reserving enough time for integration and verification. Tests and documentation should be updated alongside implementation changes, while deployment checks should provide clear evidence that verification has completed. Overall, the internship strengthened my interest in backend development and gave me a better understanding of the responsibility involved in maintaining a reliable application.

The company was good and provided a supportive environment for reaching the internship's goals. Our mentor was very helpful, was genuinely intrigued by our project, and gave us good tips and feedback that improved our work.

Date of completion: 26 / 09 / 2026

Signature of the student: Cimendur Rustem